TARGET

E-Commerce Business Case — Brazil Operations (SQL Analysis)

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Tool: SQL (MySQL dialect)

Executive Summary

Target operates in Brazil through an e-commerce marketplace covering roughly 100,000 orders placed between September 2016 and October 2018. This report analyses order volume trends, customer geography, order economics (price, freight, payments), and delivery performance using SQL, to surface actionable recommendations for the business.

1. Initial Exploration

1.1 Data Types of All Columns (customers table)

```
DESCRIBE customers;
```

name	type
customer_id	TEXT
customer_unique_id	TEXT
customer_zip_code_prefix	INTEGER
customer_city	TEXT
customer_state	TEXT

Result: customer_id, customer_unique_id, customer_city, customer_state are text; customer_zip_code_prefix is an integer.

1.2 Time Range of Orders Placed

```
SELECT MIN(order_purchase_timestamp) AS first_order_date,
       MAX(order_purchase_timestamp) AS last_order_date
FROM orders;
```

first_order_date	last_order_date
2016-09-04 21:15:19	2018-10-17 17:30:18

Result: orders span 4 September 2016 to 17 October 2018 — just over two years of data.

1.3 Count of Customer Cities & States

```
SELECT COUNT(DISTINCT customer_city) AS total_cities,
       COUNT(DISTINCT customer_state) AS total_states
FROM customers;
```

total_cities	total_states
4119	27

Result: customers ordered from 4,119 distinct cities across all 27 Brazilian states.

2. In-Depth Exploration

2.1 Year-on-Year Order Trend

```
SELECT YEAR(order_purchase_timestamp) AS order_year,
       COUNT(order_id) AS total_orders
FROM orders
GROUP BY order_year
ORDER BY order_year;
```

order_year	total_orders
2016	329
2017	45101
2018	54011

Result: orders grew from 329 (partial 2016) to 45,101 (2017) to 54,011 (2018, through October) — strong year-on-year growth.

2.2 Monthly Seasonality

```
SELECT MONTH(order_purchase_timestamp) AS order_month,
       COUNT(order_id) AS total_orders
FROM orders
GROUP BY order_month
ORDER BY order_month;
```

order_month	total_orders
1	8069
2	8508
3	9893
4	9343
5	10573
6	9412
7	10318
8	10843
9	4305
10	4959

Result: order volume is highest May-August and lowest in September-October.

2.3 Time of Day Customers Place Orders

```
SELECT
  CASE
    WHEN HOUR(order_purchase_timestamp) BETWEEN 0 AND 6 THEN 'Dawn'
    WHEN HOUR(order_purchase_timestamp) BETWEEN 7 AND 12 THEN 'Morning'
    WHEN HOUR(order_purchase_timestamp) BETWEEN 13 AND 18 THEN 'Afternoon'
    ELSE 'Night'
  END AS time_of_day,
  COUNT(order_id) AS total_orders
FROM orders
GROUP BY time_of_day
ORDER BY total_orders DESC;
```

time_of_day	total_orders
Afternoon	38135
Night	28331
Morning	27733
Dawn	5242

Result: Afternoon is the most popular ordering window (38,135 orders), followed by Night, Morning, then Dawn.

3. Evolution of E-Commerce Orders in Brazil

3.1 Month-on-Month Orders by State (sample)

```
SELECT c.customer_state AS state,
       YEAR(o.order_purchase_timestamp) AS order_year,
       MONTH(o.order_purchase_timestamp) AS order_month,
       COUNT(o.order_id) AS total_orders
FROM orders o
JOIN customers c ON o.customer_id = c.customer_id
GROUP BY state, order_year, order_month
ORDER BY state, order_year, order_month;
```

state	order_year	order_month	total_orders
AC	2017	1	2

AC	2017	2	3
AC	2017	3	2
AC	2017	4	5
AC	2017	5	8
AC	2017	6	4
AC	2017	7	5
AC	2017	8	4
AC	2017	9	5
AC	2017	10	6

First 10 rows shown (Acre state, chronological) — full result set covers all 27 states.

3.2 Customer Distribution Across States

```
SELECT customer_state AS state,
       COUNT(DISTINCT customer_unique_id) AS total_customers
FROM customers
GROUP BY state
ORDER BY total_customers DESC;
```

state	total_customers
SP	40302
RJ	12384
MG	11259
RS	5277
PR	4882
SC	3534
BA	3277
DF	2075
ES	1964
GO	1952

Result: São Paulo (SP) accounts for 40,302 customers — over 4x the next largest state (Rio de Janeiro, 12,384).

4. Impact on Economy

4.1 % Increase in Order Cost, 2017 vs 2018 (Jan-Aug)

```
SELECT
  yr17.total_2017, yr18.total_2018,
  ROUND((yr18.total_2018 - yr17.total_2017) * 100.0 / yr17.total_2017, 2) AS
pct_increase
FROM
 (SELECT SUM(p.payment_value) AS total_2017
  FROM orders o JOIN payments p ON o.order_id = p.order_id
  WHERE YEAR(o.order_purchase_timestamp) = 2017
    AND MONTH(o.order_purchase_timestamp) BETWEEN 1 AND 8) yr17,
 (SELECT SUM(p.payment_value) AS total_2018
  FROM orders o JOIN payments p ON o.order_id = p.order_id
  WHERE YEAR(o.order_purchase_timestamp) = 2018
    AND MONTH(o.order_purchase_timestamp) BETWEEN 1 AND 8) yr18;
```

total_cost_2017_jan_aug	total_cost_2018_jan_aug	pct_increase
3669022.12	8694733.84	136.98

Result: order value grew from R$3.67M to R$8.69M (Jan-Aug) — a 136.98% increase.

4.2 Total & Average Order Price by State

```
SELECT c.customer_state AS state,
       ROUND(SUM(oi.price),2) AS total_order_price,
       ROUND(AVG(oi.price),2) AS avg_order_price
```

```
FROM order_items oi
JOIN orders o ON oi.order_id = o.order_id
JOIN customers c ON o.customer_id = c.customer_id
GROUP BY state
ORDER BY total_order_price DESC;
```

state	total_order_price	avg_order_price
SP	5202955.05	109.65
RJ	1824092.67	125.12
MG	1585308.03	120.75
RS	750304.02	120.34
PR	683083.76	119.0
SC	520553.34	124.65
BA	511349.99	134.6
DF	302603.94	125.77
GO	294591.95	126.27
ES	275037.31	121.91

Result: SP drives the highest total order value (R$5.2M); Bahia (BA) has the highest average order price (R$134.60).

4.3 Total & Average Freight Value by State

```
SELECT c.customer_state AS state,
       ROUND(SUM(oi.freight_value),2) AS total_freight_value,
       ROUND(AVG(oi.freight_value),2) AS avg_freight_value
FROM order_items oi
JOIN orders o ON oi.order_id = o.order_id
JOIN customers c ON o.customer_id = c.customer_id
GROUP BY state
ORDER BY total_freight_value DESC;
```

state	total_freight_value	avg_freight_value
SP	718723.07	15.15
RJ	305589.31	20.96
MG	270853.46	20.63
RS	135522.74	21.74
PR	117851.68	20.53
BA	100156.68	26.36
SC	89660.26	21.47
PE	59449.66	32.92
GO	53114.98	22.77
DF	50625.5	21.04

Result: SP has by far the lowest average freight cost (R$15.15) thanks to proximity to distribution hubs.

5. Sales, Freight & Delivery Time Analysis

5.1 Delivery Time & Estimate Difference (single query)

```
SELECT order_id, order_purchase_timestamp,
       order_delivered_customer_date, order_estimated_delivery_date,
       DATEDIFF(order_delivered_customer_date, order_purchase_timestamp)
           AS time_to_deliver_days,
       DATEDIFF(order_delivered_customer_date, order_estimated_delivery_date)
           AS diff_estimated_delivery_days
FROM orders
WHERE order_delivered_customer_date IS NOT NULL;
```

order_id	order_purchase_timestamp	order_delivered_customer_date	order_estimated_delivery_date	time_to_deliver_days	diff_estimated_delivery_days
e481f51cbdc54678b7cc49136f2d6af7	2017-10-02 10:56:33	2017-10-10 21:25:13	2017-10-18 00:00:00	8.4	-7.1

53cdb2fc8bc7dce0b6741e2150273451	2018-07-24 20:41:37	2018-08-07 15:27:45	2018-08-13 00:00:00	13.8	-5.4
47770eb9100c2d0c44946d9cf07ec65d	2018-08-08 08:38:49	2018-08-17 18:06:29	2018-09-04 00:00:00	9.4	-17.2
949d5b44dbf5de918fe9c16f97b45f8a	2017-11-18 19:28:06	2017-12-02 00:28:42	2017-12-15 00:00:00	13.2	-13.0
ad21c59c0840e6cb83a9ceb5573f8159	2018-02-13 21:18:39	2018-02-16 18:17:02	2018-02-26 00:00:00	2.9	-9.2
a4591c265e18cb1dcee52889e2d8acc3	2017-07-09 21:57:05	2017-07-26 10:57:55	2017-08-01 00:00:00	16.5	-5.5
6514b8ad8028c9f2cc2374ded245783f	2017-05-16 13:10:30	2017-05-26 12:55:51	2017-06-07 00:00:00	10.0	-11.5
76c6e866289321a7c93b82b54852dc33	2017-01-23 18:29:09	2017-02-02 14:08:10	2017-03-06 00:00:00	9.8	-31.4
e69bfb5eb88e0ed6a785585b27e16dbf	2017-07-29 11:55:02	2017-08-16 17:14:30	2017-08-23 00:00:00	18.2	-6.3
e6ce16cb79ec1d90b1da9085a6118aeb	2017-05-16 19:41:10	2017-05-29 11:18:31	2017-06-07 00:00:00	12.7	-8.5

Negative diff_estimated_delivery_days means the order arrived before the estimated date.

5.2 Top 5 States by Average Freight Value

Highest:

state	avg_freight_value
RR	42.98
PB	42.72
RO	41.07
AC	40.07
PI	39.15

Lowest:

state	avg_freight_value
SP	15.15
PR	20.53
MG	20.63
RJ	20.96
DF	21.04

Remote northern states (RR, PB, RO, AC, PI) pay the highest freight; SP pays the least.

5.3 Top 5 States by Average Delivery Time

Slowest:

state	avg_delivery_days
RR	29.39
AP	27.19
AM	26.43
AL	24.54
PA	23.77

Fastest:

state	avg_delivery_days
SP	8.76
PR	11.99
MG	12.01
DF	12.97
SC	14.96

Roraima (RR) takes the longest to deliver (29.4 days); São Paulo (SP) is fastest (8.8 days).

5.4 Top 5 States — Fastest Delivery vs. Estimate

```
SELECT c.customer_state AS state,
       ROUND(AVG(DATEDIFF(o.order_delivered_customer_date,
o.order_estimated_delivery_date)),2)
           AS avg_diff_days
FROM orders o
JOIN customers c ON o.customer_id = c.customer_id
WHERE o.order_delivered_customer_date IS NOT NULL
GROUP BY state
ORDER BY avg_diff_days ASC
LIMIT 5;
```

state	avg_diff_days
AC	-20.08
RO	-19.4
AP	-19.06
AM	-18.85
RR	-16.59

Interesting pattern: the same remote states with the longest raw delivery times (5.3) also beat their estimated delivery date by the widest margin — Target appears to set deliberately generous delivery estimates for hard-to-reach states.

6. Payments Analysis

6.1 Month-on-Month Orders by Payment Type (sample)

```
SELECT YEAR(o.order_purchase_timestamp) AS order_year,
       MONTH(o.order_purchase_timestamp) AS order_month,
       p.payment_type,
       COUNT(DISTINCT o.order_id) AS total_orders
FROM orders o
JOIN payments p ON o.order_id = p.order_id
GROUP BY order_year, order_month, p.payment_type
ORDER BY order_year, order_month, total_orders DESC;
```

order_year	order_month	payment_type	total_orders
2016	9	credit_card	3
2016	10	credit_card	253
2016	10	UPI	63
2016	10	voucher	11
2016	10	debit_card	2
2016	12	credit_card	1
2017	1	credit_card	582
2017	1	UPI	197
2017	1	voucher	33
2017	1	debit_card	9

Credit card is consistently the dominant payment method every month.

6.2 Orders by Payment Installment Count

```
SELECT payment_installments,
       COUNT(DISTINCT order_id) AS total_orders
FROM payments
GROUP BY payment_installments
ORDER BY total_orders DESC;
```

payment_installments	total_orders
1	49060
2	12389

3	10443
4	7088
10	5315
5	5234
8	4253
6	3916
7	1623
9	644

Result: single-payment (no EMI) is most common (49,060 orders); among EMI options, 10-installment plans are the most popular.

7. Actionable Insights & Recommendations

Key Insights

- Order volume grew strongly from 2016 to 2018, with order value up ~137% (Jan-Aug 2017 vs 2018) — the Brazil operation is in a high-growth phase.
- São Paulo (SP) dominates the business: most customers, highest total order value, lowest freight cost, and fastest delivery — it is the core, most efficient market.
- Remote northern/northeastern states (RR, AP, AM, AC, RO) have the highest freight costs and longest delivery times, but Target's delivery estimates for these states are generous enough that actual delivery consistently beats the promised date.
- Afternoons and nights are the peak ordering windows; May-August is the peak ordering season, with a dip in September-October.
- Credit card is the dominant payment method every month; single-payment orders are most common, but 10-installment EMI plans are a popular financing option for larger purchases.

Recommendations

- Continue to invest in the SP hub/logistics network as the proven efficiency benchmark, and evaluate whether its low-freight, fast-delivery model can be partially replicated for neighbouring high-volume states (RJ, MG).
- Review delivery estimate accuracy for remote states — since actual delivery consistently beats the estimate by 15-20 days, tightening these estimates could improve customer experience without any operational change, by better matching customer expectations.
- Plan inventory and marketing pushes around the May-August peak season, and consider promotional campaigns in the September-October dip to smooth demand.
- Given credit card dominance, ensure card payment processing capacity and fraud checks scale with order growth; continue offering multi-installment EMI options, which are clearly used for higher-value purchases.
- Investigate freight pricing for high-cost states (RR, PB, RO, AC, PI) — either through regional fulfillment centers or renegotiated carrier rates — to reduce the cost gap with SP and improve margins on orders from these regions.